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Analytical framework for evaluating physical, biological, and fishery-related drivers of population dynamics: an application to Pacific herring

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ABSTRACT: Small pelagic fish populations experience pronounced fluctuations in abundance, likely driven by a complex interplay of physical, biological, and fishery-related factors. To better understand the mechanisms underpinning these dynamics and to disentangle these simultaneous and lagged effects without sacrificing ecological interpretability, we developed a novel randomized dynamic structural equation model (DSEM) framework and applied it to Pacific herring *Clupea pallasii* in British Columbia, Canada. We assessed the influence of 30 drivers on recruitment, age-3 weight, spawning biomass, and female body condition across 5 major herring stocks (Haida Gwaii, Prince Rupert District, Central Coast, West Coast of Vancouver Island, Strait of Georgia) from 1990 to 2024. Unlike previous studies that examined a fixed set of drivers, our approach systematically explored thousands of non-collinear driver combinations to identify emergent patterns. Biological drivers such as spring bloom date, boreal copepod biomass, combined predatory fish biomass, and harbor seal biomass consistently outperformed commonly used physical drivers such as Arctic Oscillation Index, upwelling, downwelling, and North Pacific Gyre Oscillation. Strong and consistent intrinsic effects were also observed, particularly the age-3 weight to spawning biomass and recruitment to spawning biomass pathways. Feedback mechanisms played a central role: improved female body condition enhanced recruitment, while prior spawning biomass positively influenced future body condition. These results highlight the dominant role of biological and intrinsic dynamics in regulating herring populations and demonstrate the power of the randomized DSEM framework in identifying robust, ecologically meaningful relationships. Our findings support a more nuanced, ecosystem-based approach to fisheries management under changing climatic conditions.

KEY WORDS: Dynamic structural equation modeling · *Clupea pallasii* · Ecosystem-based approach to fisheries management · EAFM · Biological and physical drivers · Population dynamics

1. INTRODUCTION

Small pelagic forage fish populations are notorious for undergoing significant fluctuations in abundance, with profound social and ecological implications that complicate fisheries management (Lluch-Belda et al. 1992, Alheit & Peck 2019). Understanding the causes of these fluctuations, which range from intrinsic factors such as stock–recruitment relationships to extrinsic influences including top-down predation and bottom-up food web dynamics, remains a central challenge in fisheries science (Peck et al. 2021, Moyano et al. 2023). Identifying the relative importance of these drivers is therefore critical for implementing an ecosystem-based approach to fisheries management, particularly in a rapidly changing environment (Pikitch

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et al. 2004). Recent advancements in analytical tools have opened new opportunities to disentangle simultaneous and lagged effects of multiple drivers on population changes (Thorson et al. 2024).

Beyond small pelagics, a mechanistic understanding of how physical, biological, and intrinsic processes jointly regulate key aspects of fish population dynamics is fundamental to ecosystem-based approaches to fisheries management (Pikitch et al. 2004). Metrics such as recruitment, spawning-stock biomass (hereafter referred to as spawning biomass, SB), individual growth (e.g. weight-at-age), and condition integrate processes across life stages and trophic levels, and are widely used to assess population status, productivity, and resilience (Hilborn & Walters 1992). Physical environmental conditions shape early survival, growth, and distribution (Cushing 1969); prey fields and primary production regulate bottom-up support (Mackas et al. 2007); predators and competitors impose top-down and density-dependent constraints (Schweigert et al. 2010); and intrinsic pathways, including stock–recruitment relationships and feedbacks between body condition and reproduction, determine how populations respond to these external pressures (Berkeley et al. 2004). Analytical frameworks capable of jointly quantifying the strength, direction, and temporal structure of these pathways therefore have broad applicability across fish species and systems, providing a general template for evaluating driver portfolios and trade-offs under ecosystem-based management (Pikitch et al. 2004, Fulton et al. 2011). Such frameworks offer an integrated, mechanistic means of identifying key drivers of population dynamics and support future predictive modeling efforts. Pacific herring *Clupea pallasii* (hereafter referred to as herring) are a widely distributed, small, pelagic forage fish found throughout the North Pacific Ocean. In British Columbia (BC), Canada, herring are managed as several distinct stocks and hold significant ecological, economic, and cultural value, especially for First Nations communities and the commercial fishing industry. Fisheries and Oceans Canada (DFO) manages herring as 5 major stocks: Prince Rupert District (PRD), Haida Gwaii (HG), Central Coast (CC), West Coast of Vancouver Island (WCVI), and Strait of Georgia (SOG) (Fig. 1; DFO 2025).

These stocks are believed to differ due to variations in spawning timing and geographic isolation of spawning sites (Cleary et al. 2018, Forrest et al. 2023, DFO 2025), a theory supported by research on genetic microsatellite variation (Beacham et al. 2008) and long-term tagging and recovery experiments (Flostrand et al. 2009). More recent genomic studies have further identified fine-scale population structure and adaptive differentiation among coastal spawning groups (Chamberlin et al. 2021, Petrou et al. 2021), supporting the persistence of spatial stock structure in BC herring.

Approximately 90% of herring reach sexual maturity by age 3, after which adults migrate from overwintering areas to coastal waters to spawn in late winter and spring (Hay 1985). These overwintering areas are generally located in offshore or deeper shelf waters adjacent to the respective spawning regions and are thought to be largely stock specific (Hay 1985, Flostrand et al. 2009). Herring populations exhibit considerable variability in abundance, driven by fluctuations in recruitment success and natural mortality occurring during both early life stages and adulthood (Ware 1991, Schweigert et al. 2010, Brosset et al. 2019, Boldt et al. 2025).

Off Canada's west coast, herring stocks have historically shown cyclical patterns of depletion, notably during the 1930s and 1960s, followed by periods of rapid recovery driven largely by reduced fishing pressure and favorable environmental conditions (Ware 1991). In the last 2 decades, declines in both abundance and weight-at-age have been documented in multiple stocks, with HG, WCVI, and CC regions experiencing closures of commercial roe herring fisheries since 2002, 2006, and 2007, respectively, due to sustained low stock levels (e.g. Cleary et al. 2018, DFO 2025). In contrast, SOG and PRD have maintained viable commercial fisheries to the present day (e.g. Cleary et al. 2018). Long-term herring stock dynamics are influenced by an array of factors, including physical, biological, and fishery-related drivers. Systematic literature mapping by Boldt et al. (2022) identified 15 such drivers affecting 21 forage fish outcomes, ranging from distribution and migration to reproduction and survival, that likely influence herring stocks through various mechanisms.

Using a multi-model approach, with univariate and multivariate statistical models, Boldt et al. (2025) investigated the impact of various physical, biological, and fishery-related factors, for which data exist, on HG herring response indicators, such as weight at age 3, recruit abundance, spawning biomass, and body condition (i.e. residuals from a double log-transformed length–weight regression). In this study, we expanded to all 5 herring stocks in order to search for common environmental drivers and to identify different influencing drivers among the 5 herring stocks. We used a unified approach in this endeavor by employing dynamic structural equation models (DSEMs; Thorson et al. 2024) to examine not only the influence of various drivers on herring response indi-

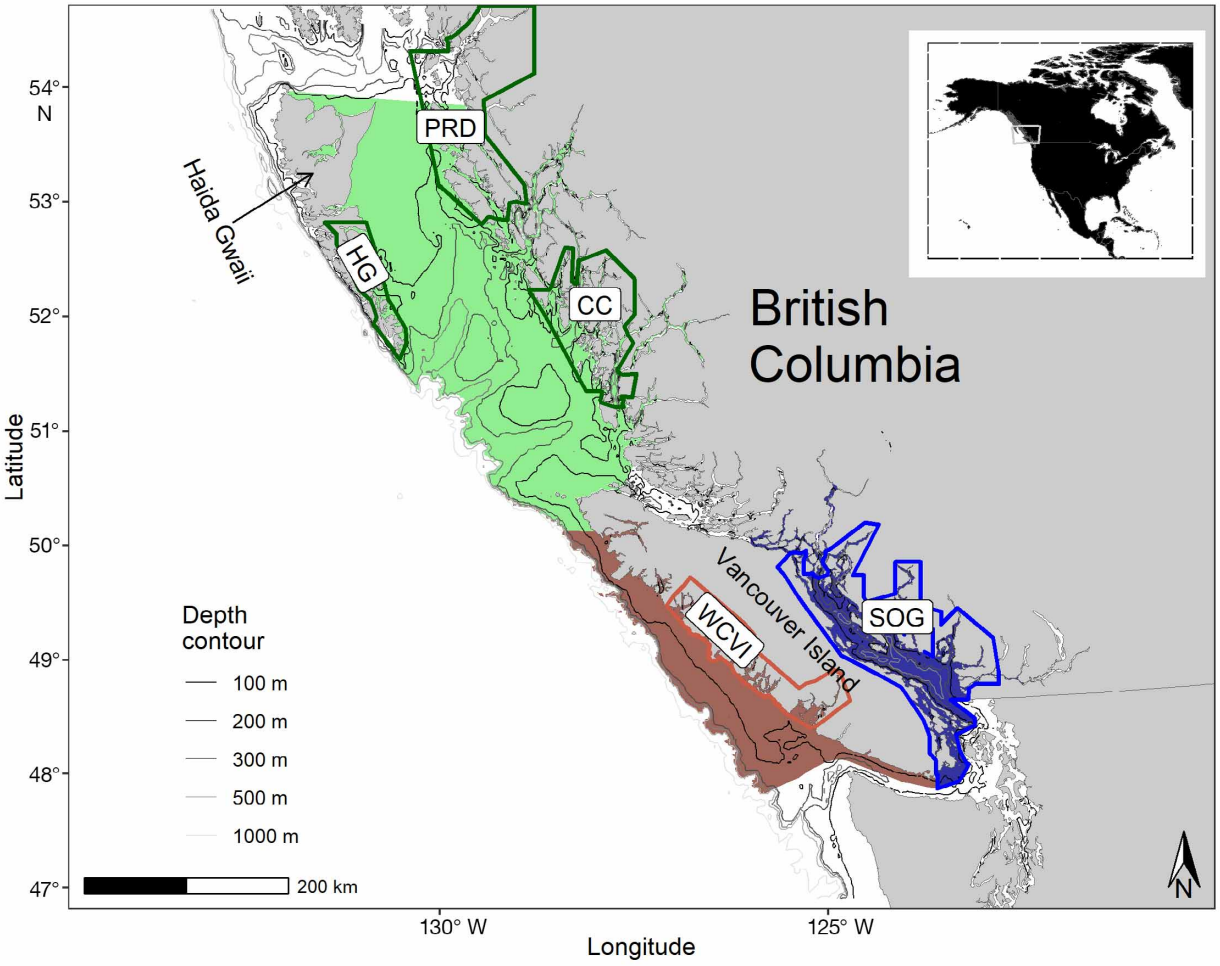


Fig. 1. Boundaries for the Pacific herring 5 major stock assessment regions in British Columbia, Canada, including Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SOG), and west coast of Vancouver Island (WCVI). Source: Fisheries and Oceans Canada. Color-shaded areas represent the feeding/mixing areas for the northern and central stocks (green) and the southern stocks (blue and brown)

cators but also the interrelationships among these response indicators.

When developing DSEMs using multiple physical, biological, and fishery-related drivers, it is crucial to address multicollinearity, a statistical phenomenon where 2 or more predictors in a multiple regression model are highly correlated. Multicollinearity can reduce the precision of the estimated coefficients, complicate the interpretation of model coefficients, and reduce the generalizability of the model (Graham 2003). Dimension-reduction techniques such as random forests, dynamic factor analysis, and principal component analysis have been employed to reduce multicollinearity (Boldt et al. 2025). However, such dimension-reduction techniques might not always capture the most informative aspects of the data regarding the underlying processes, and they can result in models where the relationships between variables are obscured or oversimplified, particularly when the original features are not only statistically significant but also carry meaningful ecological or biological interpretations (e.g. Jolliffe & Cadima 2016). To preserve interpretability while maintaining statistical rigor, we developed a novel randomized driver-sampling approach that systematically investigates non-collinear combinations of physical, biological, and fishery-related drivers within DSEMs. This approach is broadly applicable to other fish populations where multiple, potentially collinear drivers act simultaneously on recruitment, growth, spawning biomass, and condition; Pacific herring are used here as a case study to demonstrate its utility. Our objectives were to (1) identify drivers that consistently improve model performance across stocks versus those associated with poor performance; (2) disentangle intrinsic and extrinsic pathways regulating herring recruitment, growth, and spawning biomass; and (3) characterize the ecological and mechanistic implications of the optimal driver set emerging from the

optimal DSEM. Our randomized DSEM approach provides a transparent, scalable framework for small pelagic systems and delivers coastwide and regional insights into the mechanisms governing Pacific herring dynamics.

2. MATERIALS AND METHODS

2.1. Physical, biological, and fishery-related drivers

A total of 30 physical, biological, and fishery-related driver indicators were investigated, encompassing both stock-specific drivers and those common across all 5 herring stocks (Table 1). For some stock-specific drivers, regionally defined proxy indices were used and consistently assigned to relevant subsets of stocks based on spatial overlap when fully stock-specific time series were unavailable, as detailed in Table S1 in the Supplement at www.int-res.com/articles/suppl/meps15167_supp.pdf. The 8 stock-specific physical drivers, fully listed in Table 1, include seasonal sea surface temperature (SST), upwelling and downwelling (denoted as Up and Down, respectively), wind-driven transport (Wind), and sea level height (SLH). In contrast, the 7 stock-common physical drivers represent basin-scale atmospheric and oceanographic variability, including well-studied indices such as the Pacific Decadal Oscillation (PDO) and the North Pacific Gyre Oscillation (NPGO).

Biological driver indicators reflect variability in prey availability, predator pressures, and competitive dynamics (Table 1). All 6 prey-related drivers are stock-specific and include the biomass of boreal copepods (BCope), subarctic copepods (ACope), and southern copepods (SCope), total zooplankton biomass (denoted as TotalZ), and 2 indices of phytoplankton dynamics, i.e. chlorophyll *a* concentration (Chl) and spring bloom date (Bloom; the day of year on which the chlorophyll concentration exceeded 5% of the annual median chlorophyll concentration).

The 7 predator-related drivers represent the biomass of key species such as Pacific hake (Hake), combined predatory fish from groundfish surveys (denoted as GFS), humpback whales (HWhale), harbor seals (HSeal), Steller sea lions (SSLion), northern fur seals (NFurSeal), and grey whales (Table 1). Most of these predator-related drivers are stock-specific, except for NFurSeals and GreyWhale, whose biomass estimates are not available at regional scales (Table S1), and are therefore treated as stock-common. Lastly, fishery-related drivers in this study represent 2 sources of removal, i.e. herring commercial landings (Catch) and commercial bycatch (Bycatch), both stock-specific (Table 1). The roe fishery, together with other smaller harvest sectors, is included within the aggre-

Table 1. Abbreviations and full names of physical, biological (prey/primary producer and predator/competitor), and fishery-related drivers for 5 Pacific herring stocks. Stock-specific drivers are listed on the left, and drivers common to all stocks are on the right. Detailed descriptions, data sources, and references for each driver are provided in Table S1

	Stock-specific drivers		Stock-common drivers	
	Abbreviation	Full name	Abbreviation	Full name
Physical	wSST	Winter sea surface temperature	ALPI	Aleutian Low Pressure Index
	spSST	Spring sea surface temperature	AOI	Arctic Oscillation Index
	suSST	Summer sea surface temperature	MEI	Multivariate ENSO Index
	fSST	Fall sea surface temperature	NPGO	North Pacific Gyre Oscillation
	Up	Upwelling-favorable wind stress	NPI	North Pacific Index
	Down	Downwelling-favorable wind stress	PDO	Pacific Decadal Oscillation
	Wind	Wind-driven transport, Bakun index	SOI	Southern Oscillation Index
	SLH	Sea level height		
Prey	BCope	Boreal copepod biomass		
	ACope	Subarctic copepod biomass		
	SCope	Southern copepod biomass		
	TotalZ	Total zooplankton biomass		
	Bloom	Spring bloom date		
	Chl	Chlorophyll *a* concentration		
Predator	Hwhale	Humpback whale biomass	NFurSeals	Northern fur seal biomass
	Hseal	Harbor seal biomass	GreyWhale	Grey whale biomass
	SSLion	Steller sea lion biomass		
	Hake	Piscivorous Pacific hake biomass		
	GFS	Predatory groundfish survey biomass		
Fishery	Bycatch	Commercial bycatch of Pacific herring		
	Catch	Commercial landings of Pacific herring		

gated catch data used in our analyses. Thorough descriptions of these drivers, data sources, and literature references are all provided in Table S1.

2.2. Herring response indicators

We investigated 3 common herring response indicators: (1) recruitment (R), the abundance of age-2 fish, as estimated in the stock assessment model (DFO 2025; note: most herring recruit to the spawning population at age 3); (2) spawning biomass of age 3 and older (SB), as estimated in the stock assessment model (DFO 2025); and (3) weight at age 3 (Weight3), which is used because age-3 fish represent the first fully mature age class and provide a consistent measure of cohort body condition across stocks. Herring lengths and weights were measured from biological samples collected prior to spawning, as part of the stock assessment program, and used to determine Weight3. In addition, we also explored the length–weight relationship for each herring stock as a commonly studied aspect of fish biology which is often used to assess the condition, health, and well-being of fish populations. When examining the length and weight of fish, a common method is to use a double-log transformation and perform a regression analysis. The length–weight equation after log transformation is written as:

$$\ln(\text{Weight}) = a + b \times \ln(\text{Length}) + \varepsilon \quad (1)$$

where a and b are coefficients to be estimated, and ε represents the residuals or errors in the regression model. The residuals ε are used as an indicator of condition. For example, ε values in one year may be above or below average (zero), indicating good or poor condition. In this study, we specifically used the ε for females (denoted as eLWRF) as a response indicator of body condition for herring stocks, as the corresponding male indicators were found to be redundant (Boldt et al. 2025). The 30 drivers and 4 response indicators were transformed by various methods including logarithm and square root to stabilize variance and improve homoscedasticity (Fig. S1). Although herring recruitment and spawning biomass data have been available since 1951, comprehensive data on other drivers did not become available until later; for example, zooplankton data were available starting in 1983 and chlorophyll data were available starting in 2003 (Table S1). To incorporate as many indicators as possible, analyses focused on the period from 1990 to 2024; this period followed the regime shift of 1989, which resulted in notable changes in climate indices and oceanographic conditions, such as alterations in SST and ocean circulation patterns, which are critical factors influencing marine life (Hare & Mantua 2000, Ito et al. 2015).

2.3. Dynamic structural equation models

Conceptual models are widely used to communicate system understanding (e.g. Wassmann et al. 2020). The DSEM approach developed by Thorson et al. (2024) to integrate information from ecological time series provides a convenient stepping stone for translating hypotheses about simultaneous or lagged drivers into a statistical model. DSEMs can cope with missing or non-independent data, and provide the opportunity to tailor analyses based on ecological knowledge. This approach introduces a novel and computationally efficient approach to parameter estimation for non-separable DSEMs, which involves constructing a Gaussian Markov random field and fitting this using a generic interface as a generalized linear mixed model (GLMM). In this study, we constructed a DSEM for each of the 5 herring stocks (HG, PRD, CC, WCVI, and SOG) to capture the interactions among the key physical, biological, and fishery-related drivers and the herring response indicators of R, SB, Weight3, and eLWRF with predefined lagged years (Fig. 2). These time lags were specified *a priori* based on ecological knowledge of herring life history (Hay 1985) and the timing at which each response variable is measured. When spawning biomass appears as a lagged predictor in these inter-indicator pathways (e.g. SB influencing eLWRF with Lag = 3), we denote it as SBc to distinguish predictor-specific effects from contemporaneous SB used as a response variable. This 3 yr lag from SBc to eLWRF represents the time needed for offspring from a spawning event to reach age 3, when female condition is evaluated and when herring join the spawning stock. Fishery-related drivers were linked to spawning biomass using a 3 yr lag to reflect SB as ages 3 and older in the stock assessment, such that fishing impacts primarily influence future spawning cohorts rather than same-year SB estimates. The DSEM approach explicitly quantifies the strength, direction, and temporal structure of physical, biological, and fishery-related pathways affecting herring populations. This modeling framework provides an integrated and mechanistic understanding of the drivers regulating herring dynamics, offering a foundation for more informed ecosystem-based management and future predictive modeling efforts.

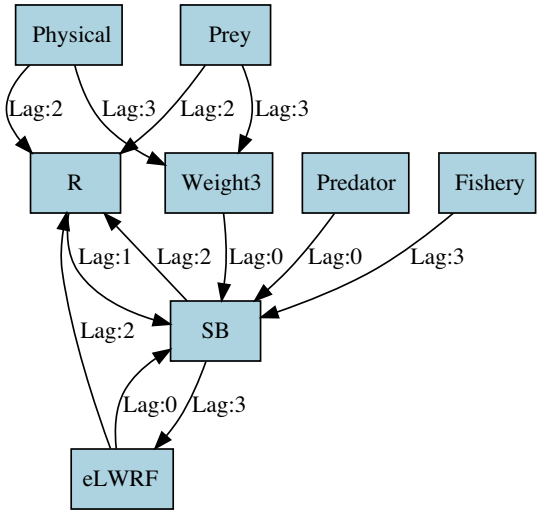


Fig. 2. Pathways along with the lagged (Lag) number of years among drivers of physical, biological (prey and predator), and fishery drivers, and the 4 Pacific herring response indicators, including recruit abundance (R), weight at age 3 (Weight3), spawning biomass (SB), and linear regressed residuals of double-log transformed length and weight relationship for females (eLWRF) as measures of herring body condition

To explore the relative influence of physical, biological, and fishery-related drivers on herring dynamics, we constructed and compared DSEM scenarios using systematically sampled combinations of drivers. In DSEMs, managing multicollinearity among predictors is crucial to ensure stable and reliable model estimates. Correlation coefficients exceeding 0.8 ($|r| > 0.8$) are typically indicative of strong collinearity, which can inflate standard errors and distort coefficient estimates, undermining model reliability (Graham 2003). Among the 5 herring stocks, such high correlations were observed only among a limited subset of drivers, most notably within marine mammal biomass time series (e.g. HWhale, SSLion, and NFurSeals) and between SOI and MEI (Fig. S2).

To prevent collinear drivers from being selected together in a single model scenario, all candidate drivers were grouped into 6 clusters based on pairwise correlation structure and thematic similarity (Table 2). Drivers exhibiting strong positive or negative correlations ($|r| > 0.8$) were assigned to the same group to ensure they would not co-occur in any DSEM scenario. For example, SOI and MEI were placed in the same group together, as were HWhale, SSLion, and NFurSeals, due to their consistently high correlations across stocks. Drivers with moderate correlations ($0.6 \leq |r| < 0.8$; Fig. S2) were also grouped together to reduce the chance of including moderately collinear drivers in the same DSEM scenario. In addition to correlation patterns, thematic coherence, such as grouping together physical drivers, predator biomass, zooplankton drivers, and seasonal SST, was incorporated to enhance ecological interpretability within each cluster.

To construct each DSEM scenario, we randomly sampled 1 driver from each of the 6 groups (Table 2). This sampling strategy ensured that (1) each driver had an equal probability of selection, (2) sampling was balanced across groups, (3) multicollinearity was minimized by restricting correlated drivers from appearing, and (4) sufficient variation was retained across scenarios to support robust inference. This design yielded a total of 15 625 possible driver combinations, representing all unique sets formed by selecting 1 driver from each of the 6 groups; this total reflects the full combinatorial space of driver sets and is independent of the number of stocks or response indicators. While evaluating all combinations would be computationally intensive and unnecessary, we estimated the required sample size for scenario coverage using the formula

$$n = \frac{N}{1 + N \times e^2} \quad (2)$$

where N is the total number of combinations and e is the desired margin of error. With a 1% margin of error, the required number of sampled scenarios was calculated to be 6098. The same set of 6098 scenarios was applied identically to each of the 5 herring stocks, ensuring consistent model structure and allowing direct comparison across stocks. Analyses were conducted in R (R Core Team 2024) using the package 'dsem' following the implementation described by Thorson et al. (2024).

2.4. Comparisons of DSEM scenarios

To compare the performance of the 6098 DSEM scenarios, we used Akaike's information criterion (AIC),

Table 2. Six groups of drivers (full names are given in Table 1 and Table S1) from each of which 1 sample was randomly selected to construct the dynamic structural equation models

Group	Drivers included within each group				
1	AOI	Chl	Bloom	Up	NPI
2	wSST	spSST	suSST	fSST	NPGO
3	SLH	Wind	ALPI	Down	BCope
4	Bycatch	GreyWhale	ACope	TotalZ	GFS
5	SOI	MEI	PDO	Hake	SCope
6	HWhale	SSLion	NFurSeals	HSeal	Catch

a widely applied metric for model selection that balances model fit and complexity (Burnham & Anderson 2002). AIC was calculated for each DSEM scenario. In addition, we computed ΔAIC to facilitate model comparisons across the 5 herring stocks. ΔAIC is defined as the difference between the AIC of a given model and that of the best-performing model: $\Delta AIC = AIC_i - AIC_{min}$, where AIC_i is the AIC of the model i and AIC_{min} is the minimum AIC among all scenarios. According to the 'AIC Rule' of Burnham & Anderson (2002), models with $\Delta AIC \leq 2$ are considered to have equivalent support, while models with $\Delta AIC > 10$ are regarded as substantially inferior. Because all stocks were evaluated using the exact same 6098 scenarios, ΔAIC values are directly comparable among stocks.

In our analysis, we focused on identifying drivers that consistently appeared in models with the lowest 2.5th percentile or the highest 97.5th percentile AIC values. Drivers associated with the lowest AIC values were considered potentially beneficial, as they contributed to more effective model performance. In contrast, drivers frequently appearing in the highest-AIC scenarios were interpreted as contributing less explanatory value under the DSEM framework, as their inclusion was associated with comparatively poorer model fit. This approach allowed us to assess both the best- and worst-performing models, providing a more comprehensive understanding of how different drivers influence model quality and enhancing the robustness of our statistical evaluation.

We also assessed the importance of each pathway across all DSEM scenarios using a pathway importance scoring system. This scoring system integrates 3 key factors: magnitude, stability, and significance for each pathway coefficient. For magnitude, we used the absolute value of the mean coefficient, which ensures that strong effects, whether positive or negative, are considered equally impactful. Stability is gauged by the inverse of the standard deviation of the coefficient estimates across different scenarios. A lower standard deviation indicates higher stability, implying that the effect of the pathway is consistent across models. For significance, we transformed the p-values using exponential decay, i.e. exp(−average p-value), with smaller p-values creating higher values of the significance score, indicating greater statistical significance. This transformation makes the p-values more interpretable by emphasizing smaller p-values. The overall importance score for each pathway was then calculated as the product of these 3 components: Importance Score = Magnitude × Stability × Significance Score. This scoring system allowed us to directly compare the relative importance of each pathway, highlighting those that consistently contribute to model performance across a variety of scenarios.

2.5. Pathway analysis and Bayesian inference

For the best-performing DSEM, identified as the model with the lowest overall AIC summed across all 5 herring stocks, we conducted pathway analysis to examine the strength and variability of individual pathways in detail. For this best-performing DSEM, Bayesian inference was also employed to further assess the uncertainty associated with pathway estimates and to evaluate the overall validity of the model. The R package 'tmbstan' was used to run Bayesian inference using Markov chain Monte Carlo (MCMC) simulations on the DSEM (Monnahan & Kristensen 2018). The use of MCMC allows for Bayesian inference, providing a way to obtain full posterior distributions of the model parameters rather than just point estimates. After running MCMC simulations, convergence of the chains was checked based on various diagnostic plots (e.g. trace plots: visually assessing the mixing and convergence of the chains) and statistics. We examined 2 statistics, including effective sample size (ESS) and the R-hat or Gelman-Rubin statistic (Gelman & Rubin 1992, Vehtari et al. 2021). ESS indicates how many independent-like samples are in the chain. Larger values are better, as they suggest that the sampling algorithm is effectively exploring the parameter space. An R-hat value close to 1 suggests that the chains have converged to a stable solution. We specifically plotted the posterior distributions (showing parameter estimates and their credible intervals) to provide both visual and numeric representations of the probability distributions of the pathway coefficients. They help in understanding the uncertainty and possible values the pathway coefficients can take.

3. RESULTS

3.1. Pathway coefficients of DSEM scenarios

The mean coefficients of the 57 pathways derived from the 6098 DSEM scenarios varied across the 5 herring stocks (HG, PRD, CC, WCVI, SOG), highlighting spatial heterogeneity in both the magnitude and direction of driver effects (Fig. 3). Intrinsic stock dynamics pathways generally demonstrated strong and uniform effects across all stocks. The R_to_SB

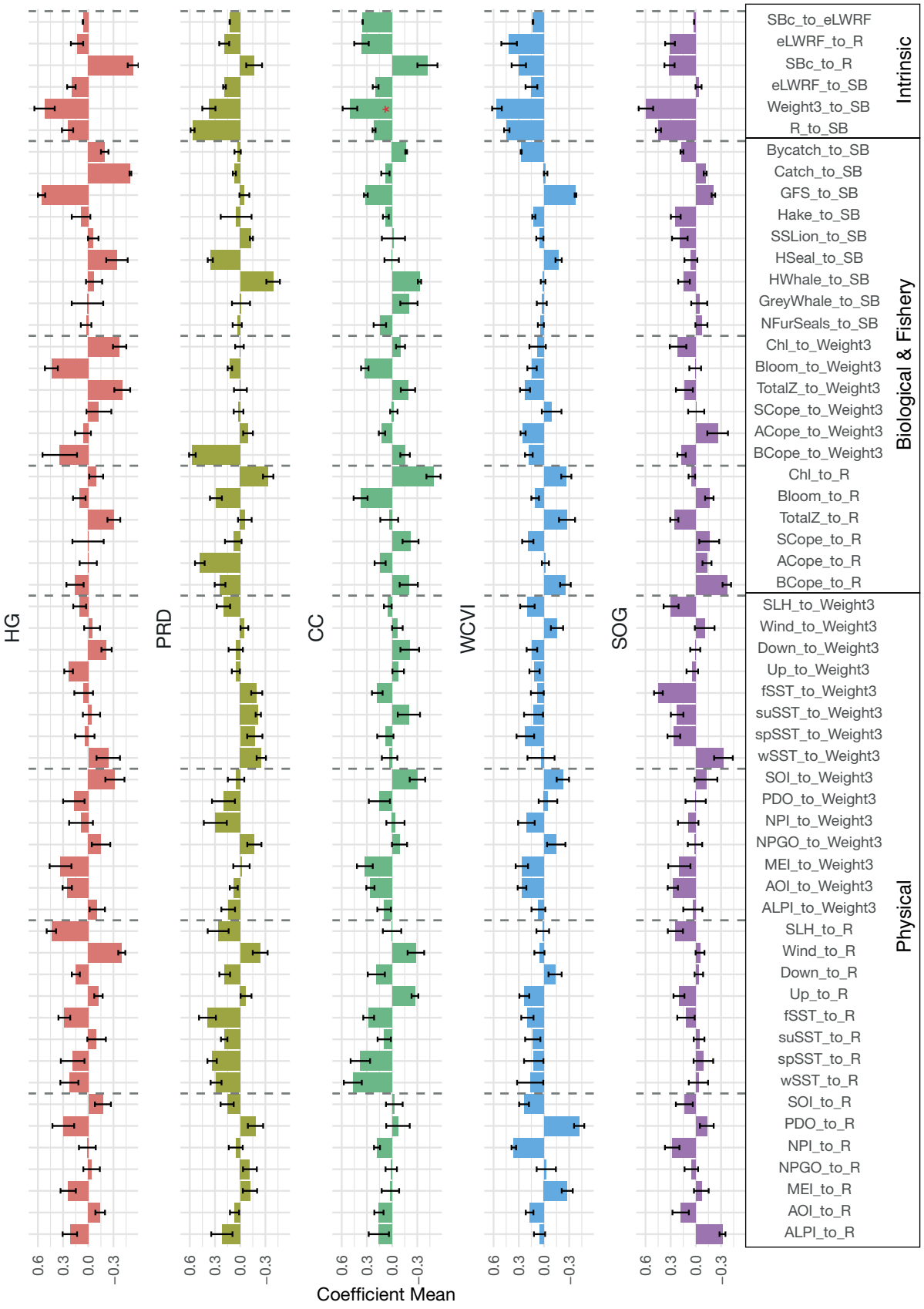


Fig. 3. Mean and standard error of coefficient for each of the 57 pathways across 5 Pacific herring stocks—Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), West Coast of Vancouver Island (WCVI), and Strait of Georgia (SOG)—derived from all the dynamic structural equation modeling (DSEM) scenarios for the period 1990–2024. The names and descriptions of the drivers and herring response indicators associated with these pathways are provided in Table S1

and Weight3_to_SB pathways were consistently positive and relatively strong, with moderate standard errors, supporting the biological expectation that higher recruitment and larger body size at recruitment contribute to increased spawning biomass. Similarly, the eLWRF_to_R and SBc_to_eLWRF pathways were consistently positive across all stocks, highlighting feedbacks in which improved female body condition led to higher recruitment 2 yr later, while greater spawning biomass enhanced individual body condition 3 yr later.

Physical drivers, particularly seasonal SST, showed both consistent and stock-specific influences (Fig. 3). With the exception of SOG, stocks HG, PRD, CC, and WCVI generally exhibited strong, positive relationships between seasonal SST (wSST, spSST, suSST, fSST) and R. In contrast, SST effects on Weight3 varied regionally, with negative effects in PRD, positive effects in WCVI and SOG, and more variable responses in HG and CC, suggesting that SST influences on juvenile growth are regionally dependent and potentially shaped by local food web dynamics, prey phenology, and competitive interactions.

Biological and fishery-related pathways also varied across regions. TotalZ_to_R was negative in HG and WCVI, but positive in SOG, whereas Chl_to_R was negative in PRD, CC, and WCVI (Fig. 3). In contrast, Bloom_to_R was positive in HG, PRD, and CC, suggesting that earlier phytoplankton blooms may enhance recruitment success. For Weight3-related pathways, Bloom_to_Weight3 was positive for most stocks, indicating beneficial effects of earlier blooms on juvenile growth. Predator_to_SB pathways were highly stock-specific: HWhale showed strong negative effects in PRD and CC, HSeal was negative in HG and WCVI but positive in PRD, and GFS had negative effects in WCVI and SOG but positive effects in HG and CC (Fig. 3). Among fishery-related drivers, Catch generally exhibited negative effects on SB, whereas Bycatch was positively associated with SB in WCVI and SOG but negatively associated in HG and CC.

3.2. Pathway importance scores of DSEM scenarios

The pathway importance scores revealed consistent patterns across the 5 herring stocks, highlighting key ecological processes regulating stock dynamics (Fig. 4). Intrinsic stock dynamics pathways exhibited the highest importance scores overall. In particular, the SBc_to_eLWRF pathway consistently ranked among the most important across all stocks, underscoring a robust feedback mechanism whereby greater spawning biomass enhances individual body condition 3 yr later. The R_to_SB and Weight3_to_SB pathways also displayed high importance, reflecting the central roles of recruitment strength and individual size in determining future spawning biomass.

Among physical drivers, seasonal SST pathways affecting recruitment tended to be more important in PRD and CC than in the other stocks. In contrast, SST pathways influencing Weight3 were more influential in PRD and SOG than in the other stocks. Basin-scale driver pathways, notably NPI_to_R showed high importance for WCVI and moderate importance for CC and SOG.

In contrast, biological pathways tended to exert stronger and more consistent influence than physical drivers. Bottom-up pathways such as BCope_to_R, ACope_to_R, Chl_to_R, and BCope_to_Weight3 exhibited moderate to high importance scores in most stocks, reaffirming the critical role of prey availability in supporting recruitment and early growth. Predator-related pathways showed greater regional variation. Among these, the GFS_to_SB pathway stood out with particularly high importance in most stocks, surpassing all other predator pathways except in CC, where HWhale_to_SB was more influential (Fig. 4).

Fishery-related pathways also exhibited spatial variability. Catch_to_SB had a very high importance score in HG, but only low to moderate scores in other stocks, indicating regional differences in the impact of commercial fisheries on spawning biomass. In contrast, Bycatch_to_SB showed high importance in CC, WCVI, and SOG, while its influence was low to moderate in the remaining stocks.

3.3. Identifying important drivers through DSEM scenarios

Drivers that occurred most frequently in the lowest 2.5th percentile of AIC values (Table 3) were considered key contributors to model performance. However, their influence varied across stocks, particularly between the 2 northern stocks (HG and PRD) and the central and southern stocks (CC, WCVI, and SOG). For instance, BCope and SCope were the most frequently selected drivers in HG and PRD, ranking first and second, but ranked considerably lower in the remaining stocks. In contrast, GFS was the most frequently selected driver in the best-performing scenarios for CC, WCVI, and SOG, but it ranked only sixth in HG and PRD, indicating a clear geographic distinction in its relative importance. While Hake did

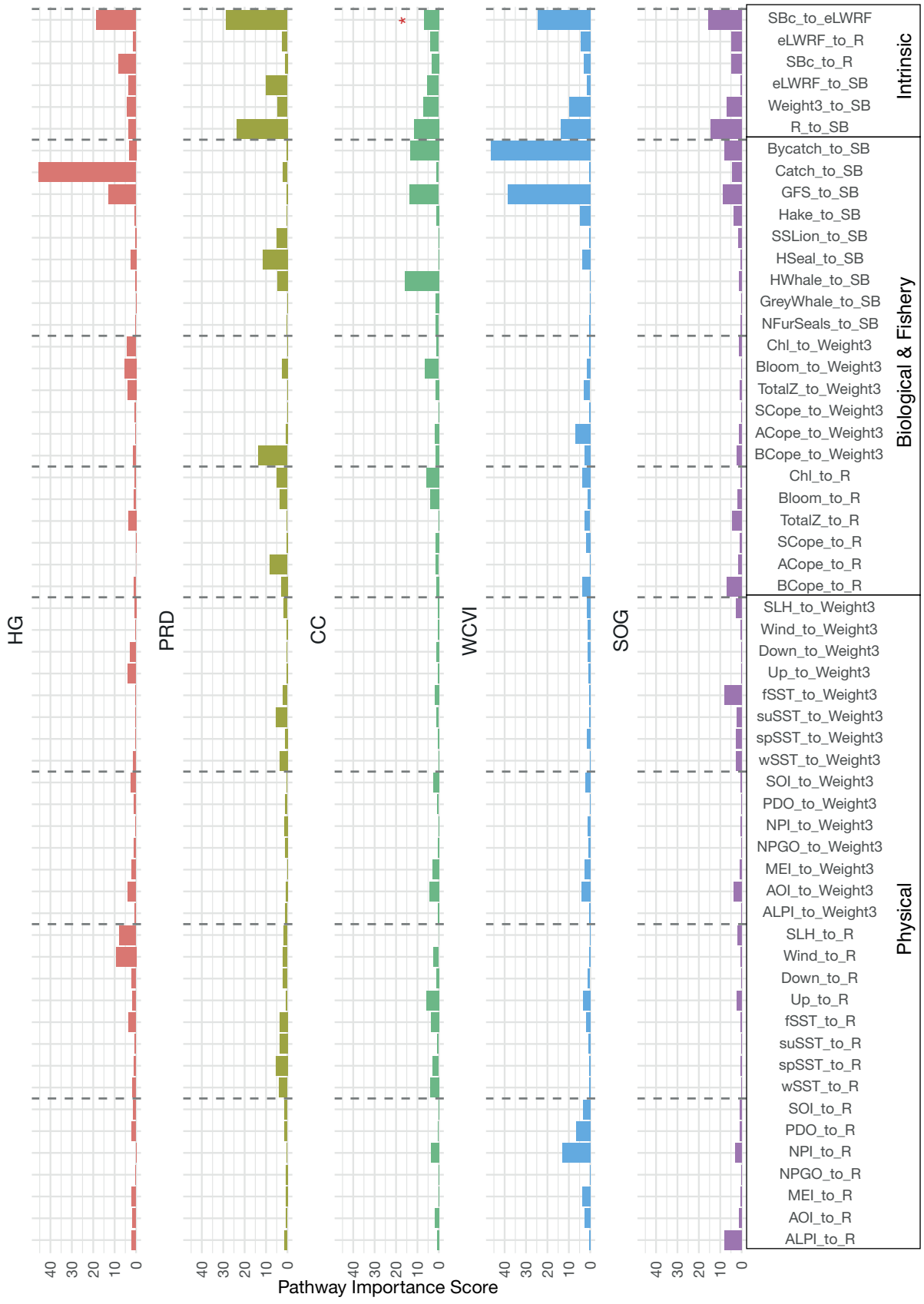


Fig. 4. Importance scores for each of the 57 pathways across 5 Pacific herring stocks — Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), West Coast of Vancouver Island (WCVI), and Strait of Georgia (SOG) — derived from all DSEM scenarios for the period 1990–2024. Each pathway is represented by a single aggregated importance score summarizing its contribution across the full set of scenarios. The names and descriptions of the drivers and herring response indicators associated with these pathways are provided in Table S1. The red asterisk (*) indicates extreme values exceeding 5 times the 95th percentile, which have been log-transformed for improved visualization

appear among the top-ranking drivers in terms of occurrences, it was associated with substantially higher AIC values (ΔAIC > 10) in HG and PRD and was therefore considered a markedly weaker predictor of herring dynamics.

Despite these regional contrasts, several drivers showed consistent importance across all 5 stocks. Notably, Chl and Bloom, both originating from the same group and performing equally well, were consistently ranked third or fourth, highlighting their broad ecological relevance in regulating herring dynamics.

Altogether, 7 drivers emerged as the most influential across the full set of DSEM scenarios. These included Bloom and Chl from Group 1, suSST from Group 2, BCope from Group 3, GFS from Group 4, and HSeal and Catch from Group 6 (Table 2). To avoid redundancy, we excluded Chl in the following analyses because both Bloom and Chl originate from the same group and performed equally well. In contrast, although HSeal and Catch also originate from the same group, they represent distinct ecological processes, namely predation pressure and fishery removal, and their inclusion together improved model performance, so both drivers were retained in the optimal configuration. However, we note that Bloom (phenology) and Chl (abundance) represent distinct ecological mechanisms, and their exclusion here should not be interpreted as implying biological substitutability. Each driver represents a distinct thematic group, and together they form the optimal DSEM configuration. This set of key drivers captures a suite of critical ecological processes, including productivity, thermal condition, prey availability, predation, and fishing pressure, that underpin herring dynamics across the coastwide study regions.

Conversely, we identified poor-performing drivers that most frequently occurred in the DSEM scenarios with the highest AIC values across all 5 herring stocks (Table 4). The AIC thresholds defining poor model performance (i.e. above the 97.5th percentile) were relatively narrow, ranging from 1.6 in SOG to 3.37 in PRD, indicating similarly poor performance across the 5 herring stocks. Among these poor-performing drivers, GreyWhale ranked first in frequency of occurrence in the highest-AIC scenarios for all 5 herring stocks. AOI was the second most frequent in HG and PRD and ranked third in SOG. The downwelling index (Down) also featured prominently among the worst-performing DSEM scenarios, placing second in SOG, third in HG and WCVI, and fourth in PRD and CC. MEI occurred with nearly the same frequency as Down, while Up was frequently present in the poorest-performing scenarios for CC and WCVI (second), PRD (third) and HG (fourth), but did not appear among the poor-performing drivers in SOG.

To mirror the identification of the 6 best-performing drivers, we also recognized 6 consistently poor-performing drivers across all DSEM scenarios. To further evaluate the robustness of the original optimal DSEM configuration, each of the 6 top-ranked drivers was individually substituted with another driver generally from the same group that was frequently associated with poor model performance (Table 5). The original optimal configuration produced a cumulative AIC of 5149.08 across all 5 stocks. Substituting Bloom with AOI substantially increased the cumulative AIC to 5350.15, indicating a significant decline in model performance. Replacing suSST with NPGO resulted in only a minor increase in AIC, which was not statistically significant.

For the first 3 driver substitutions, the 2 northern stocks (HG and PRD) consistently showed lower AIC values than the central or southern stocks (CC, WCVI, and SOG). When BCope was replaced by Down in the fourth driver set, the 2 northern stocks experienced a marked increase in AIC, while the other 3 stocks showed AIC increases of less than 10. In this case, ΔAIC values among the 5 stocks were all below 5, indicating that these models had weaker support than the best-performing model but were not markedly inferior; in contrast, substituting GFS with GreyWhale resulted in a significant deterioration in model performance, increasing the cumulative AIC from 5457.10 to 5644.09. Similarly, replacing HSeal with HWhale led to a further increase in cumulative AIC to 5716.74. Again, substituting HWhale with MEI and GreyWhale with Up increased AIC values but the increases were not statistically significant.

These substitution tests underscore the critical role of the original 6-driver configuration. Replacing any of the top-performing drivers with less effective alternatives from the same thematic group consistently degraded model performance, reinforcing their importance in explaining herring stock dynamics.

3.4. Pathway coefficients and structural relationships in the optimal DSEM configuration

The estimated simultaneous and lagged effects from the optimal DSEM revealed several robust and significant relationships that structured herring population dynamics across the 5 major stocks, as evidenced by both the pathway coefficients and their associated posterior uncertainty (Figs. 5 & 6).

Table 3. Drivers within the 5 Pacific Herring stocks (HG: Haida Gwaii; PRD: Prince Rupert District; CC: Central Coast; WCVI: West Coast of Vancouver Island; and SOG: Strait of Georgia) that most frequently occurred (count indicates number of occurrences) in high-performing dynamic structural equation models (DSEMs) for the period 1990–2024. Included models are those with Akaike's information criterion (AIC) values within the lowest 2.5th percentile of all DSEM scenarios for each stock. Drivers with consistently high counts and low average AIC (Avg_AIC) across all stocks are highlighted using distinct colors to aid visual comparison; the same color denotes the same driver across all stocks, and colors are supplementary to the numerical values and rankings. Full names and descriptions of the drivers are provided in Table S1

HG			PRD			CC			WCVI			SOG		
Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC
BCope	125	1001.91	BCope	127	1002.65	GFS	153	1046.09	GFS	153	1045.53	GFS	137	1037.49
SCope	100	999.58	SCope	99	999.94	HSeal	90	1045.36	HSeal	84	1044.37	SLH	118	1037.08
Chl	78	1003.65	Chl	79	1004.58	Bloom	77	1046.00	Chl	77	1045.59	Chl	88	1038.46
Bloom	72	1003.75	Bloom	71	1003.93	Chl	76	1046.19	Bloom	76	1045.46	Bloom	63	1038.15
HSeal	71	1004.71	HSeal	66	1004.45	Hake	75	1044.43	Hake	75	1043.78	HSeal	53	1036.03
GFS	57	1003.95	GFS	59	1005.34	ALPI	42	1045.69	ALPI	42	1045.88	Hake	52	1036.58
ACope	48	1002.31	ACope	45	1001.69	BCope	35	1045.42	SLH	38	1043.15	suSST	42	1039.60
TotalZ	40	1004.27	fSST	44	1005.47	suSST	33	1045.17	BCope	35	1045.40	Catch	35	1039.11
Hake	40	1011.19	Hake	43	1012.14	SLH	32	1044.33	suSST	34	1044.78	fSST	33	1038.80
fSST	39	1004.61	TotalZ	40	1004.54	wSST	32	1047.24	fSST	33	1045.50	spSST	31	1038.58
suSST	36	1003.09	suSST	34	1004.71	fSST	31	1045.57	NPGO	30	1045.87	PDO	28	1039.87
spSST	29	1005.07	NPGO	32	1003.39	NPGO	29	1046.01	wSST	29	1045.99	SCope	27	1039.16
NPGO	28	1002.17	Catch	24	1006.20	spSST	28	1046.53	Catch	29	1046.60	NPGO	26	1038.11
Catch	23	1005.73	NFurSeals	23	1003.13	Catch	25	1046.40	SCope	29	1046.76	NFurSeals	24	1039.77
NFurSeals	22	1001.56	wSST	23	1005.25	SCope	24	1046.34	spSST	27	1045.61	MEI	24	1041.21
wSST	21	1004.68	spSST	20	1002.65	Wind	22	1047.64	Down	20	1048.72	SOI	22	1036.95
SSLion	19	1001.99	SSLion	20	1003.65	Down	22	1048.97	Wind	18	1046.41	wSST	21	1035.59
HWhale	18	1003.25	HWhale	20	1004.77	SOI	20	1047.83	SOI	18	1046.55	HWhale	21	1042.18
ALPI	15	1012.83	ALPI	15	1013.15	PDO	17	1048.09	NFurSeals	17	1046.75	ALPI	21	1044.33
SLH	8	1011.69	Bycatch	9	1012.23	MEI	17	1049.04	MEI	16	1048.55	SSLion	20	1037.95
Bycatch	7	1010.60	SLH	8	1012.62	NFurSeals	16	1046.96	PDO	15	1047.42	Bycatch	13	1046.22
PDO	6	1014.74	SOI	5	1014.98	SSLion	13	1047.88	SSLion	14	1047.04	BCope	7	1043.71

Table 4. Drivers within the 5 Pacific herring stocks (abbreviations as in Table 3) that most frequently occurred (count indicates number of occurrences) in poor-performing dynamic structural equation models (DSEMs) for the period 1990–2024. Included models are those with Akaike's information criterion (AIC) values within the highest 97.5th percentile of all DSEM scenarios for each stock. Drivers with consistently high counts and low average AIC (Avg_AIC) across all stocks are highlighted using distinct colors to aid visual comparison; the same color denotes the same driver across all stocks, and colors are supplementary to the numerical values and rankings. Full names and descriptions of the drivers are provided in Table S1

HG			PRD			CC			WCVI			SOG		
Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC	Selected Drivers	Count	Avg_AIC
GreyWhale	153	1142.68	GreyWhale	153	1143.30	GreyWhale	119	1144.23	GreyWhale	125	1143.13	GreyWhale	82	1145.41
AOI	68	1142.82	AOI	66	1143.54	Up	86	1144.14	Up	78	1143.29	Down	81	1145.44
Down	66	1143.34	Up	66	1143.83	MEI	76	1144.44	Down	78	1143.49	AOI	66	1145.33
Up	65	1143.09	Down	65	1144.11	Down	72	1144.52	MEI	75	1143.51	MEI	66	1146.14
MEI	57	1143.25	MEI	62	1143.747221	Wind	65	1144.07	AOI	68	1143.14	Wind	62	1145.21
PDO	55	1142.06	PDO	55	1142.76	AOI	62	1144.22	Wind	62	1143.07	SSLion	56	1145.14
Wind	53	1143.32	Wind	55	1143.85	HWhale	56	1144.10	NFurSeals	48	1143.09	HWhale	53	1145.42
NPGO	45	1143.19	HWhale	46	1143.12	spSST	47	1144.12	HWhale	48	1143.30	wSST	50	1145.35
SSLion	44	1142.85	SSLion	45	1143.45	SSLion	44	1144.40	spSST	44	1143.17	spSST	49	1145.40
wSST	44	1143	NPGO	40	1143.33	NFurSeals	41	1144.21	SSLion	41	1143.52	ACope	44	1145.21
NFurSeals	43	1142.82	NFurSeals	39	1143.97	NPGO	40	1144.23	NPGO	40	1143.44	NPGO	43	1145.30
SOI	41	1142.71	wSST	37	1143.80	wSST	37	1144.58	PDO	38	1143.04	NFurSeals	40	1145.42
HWhale	40	1142.95	SOI	36	1143.34	PDO	36	1144.05	wSST	38	1143.56	PDO	34	1144.68
Catch	26	1141.74	spSST	31	1143.89	SOI	31	1143.86	SOI	29	1142.99	SOI	29	1144.62
spSST	25	1142.08	suSST	26	1142.58	TotalZ	17	1143.48	suSST	17	1142.56	TotalZ	27	1145.11
fSST	21	1142.19	Catch	23	1142.22	suSST	17	1143.75	ACope	17	1143.18	SCope	24	1144.66
NPI	20	1140.84	NPI	21	1140.84	ACope	17	1143.98	Catch	16	1141.86	BCope	10	1144.71
suSST	18	1141.18	ALPI	19	1140.73	fSST	12	1142.83	fSST	14	1141.76	suSST	9	1144.61
SLH	17	1140.15	fSST	19	1142.25	Catch	12	1142.84	SCope	11	1141.45	Catch	4	1144.64
ALPI	17	1140.64	SLH	14	1140.84	SCope	10	1142.68	TotalZ	11	1143.23	fSST	2	1144.55
						SLH	6	1142.39	BCope	7	1141.40			
						BCope	6	1142.64	NPI	7	1141.50			
						NPI	5	1142.51	ALPI	6	1141.36			
						ALPI	4	1142.36						

Table 5. Akaike's information criterion (AIC) and ΔAIC values for dynamic structural equation models (DSEMs) of 5 Pacific Herring stocks (abbreviations as in Table 3) under sequentially modified driver sets. The top-performing driver set for each stock (first row) was progressively altered by substituting in AOI, NPGO, Down, GreyWhale/Up, and HWhale/MEI. ΔAIC was calculated individually for each driver set as the difference between the AIC of a given stock and the lowest AIC observed among the 5 stocks for that specific driver set. Full names and descriptions of the drivers are provided in Table S1

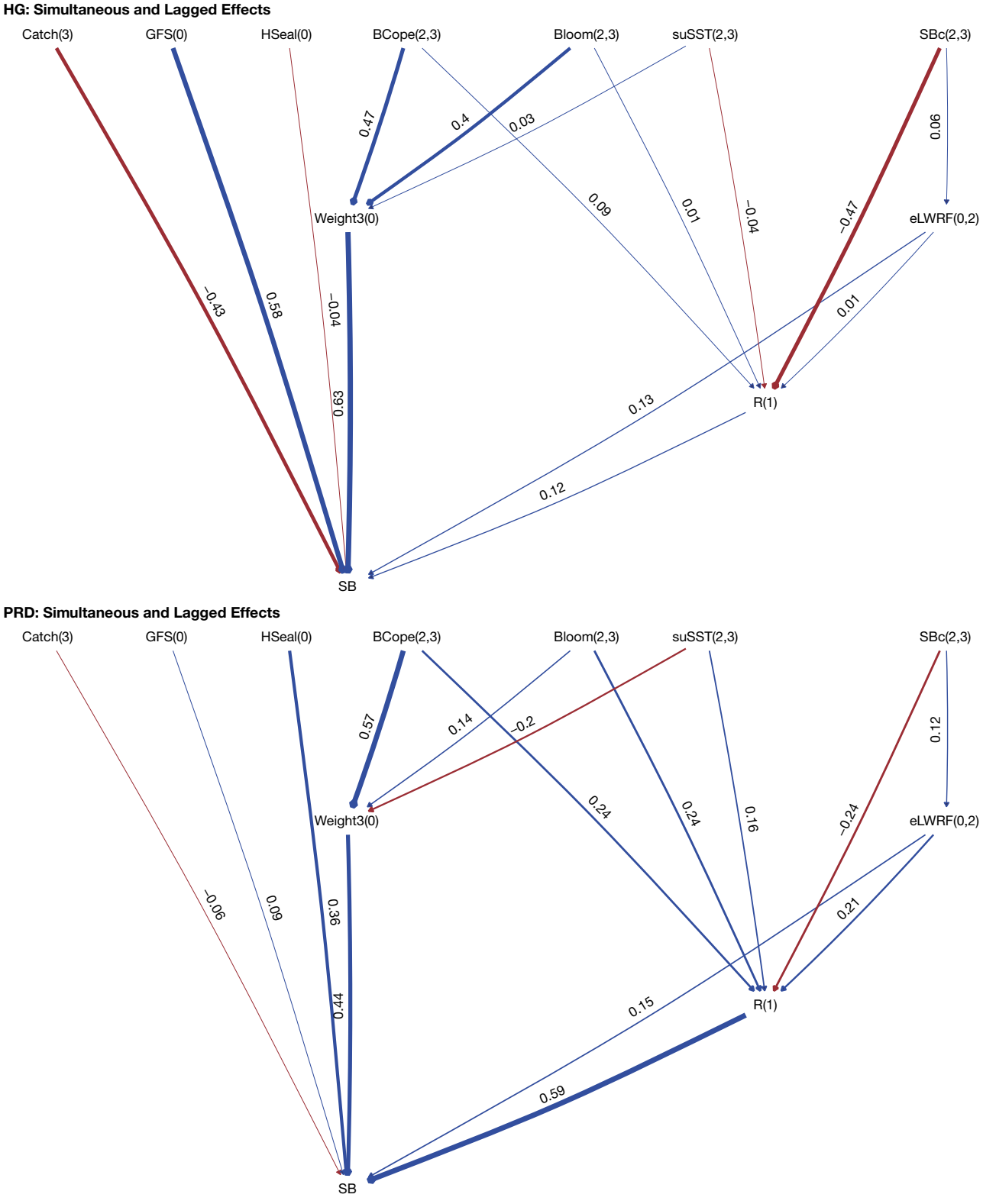
Drivers	AIC					AIC sum across stocks	ΔAIC				
	HG	PRD	CC	WCVI	SOG		HG	PRD	CC	WCVI	SOG
Bloom, suSST, BCope, GFS, HSeal, Catch	1008.54	1010.81	1042.02	1042.01	1045.71	5149.08	0	2.28	33.48	33.47	37.17
AOI, suSST, BCope, GFS, HSeal, Catch	1048.93	1051.21	1082.33	1082.10	1085.58	5350.15	0	2.28	33.40	33.17	36.65
AOI, NPGO, BCope, GFS, HSeal, Catch	1054.73	1054.10	1085.24	1085.33	1088.58	5367.98	0.64	0	31.15	31.23	34.48
AOI, NPGO, Down, GFS, HSeal, Catch	1092.07	1091.55	1090.87	1091.17	1091.45	5457.10	1.21	0.68	0	0.30	0.58
AOI, NPGO, Down, GreyWhale, HSeal, Catch	1129.63	1128.89	1128.53	1128.33	1128.70	5644.09	1.31	0.56	0.20	0	0.38
AOI, NPGO, Down, GreyWhale, HWhale, Catch	1144.01	1143.61	1143.33	1142.58	1143.21	5716.74	1.43	1.03	0.75	0	0.63
AOI, NPGO, Down, GreyWhale, MEI, Catch	1145.57	1145.38	1144.81	1144.30	1144.89	5724.94	1.27	1.09	0.52	0	0.59
AOI, NPGO, Down, Up, MEI, Catch	1147.57	1147.32	1146.88	1146.07	1146.66	5734.50	1.50	1.24	0.80	0	0.59

Fig. 5 illustrates the direction, strength, and time lag of each pathway, with arrow thickness representing the magnitude of effect and colors denoting direction (blue = positive, red = negative effect). Because Fig. 5 is based on a single optimized run, it provides a clear visualization of the structural relationships included in the best-performing DSEM. In contrast, Fig. 6 summarizes cross-stock patterns by presenting posterior distributions of pathway coefficients, standard errors, and significance levels derived from the full MCMC sampling. Significance was evaluated using Wald *z*-tests based on the estimated coefficients and their standard errors from the fitted DSEM. Minor differences between Figs. 5 & 6 are expected, as the former reflects point estimates, whereas the latter captures full posterior uncertainty.

Among all modeled relationships, the intrinsic pathways consistently exhibited the strongest effects. Notably, the positive impact of Weight3 on SB was significant across all 5 stocks ($p < 0.001$), indicating that heavier weight at age 3 reliably contributes to increased spawning biomass (Fig. 6). Similarly, the R_to_SB pathway was significant in all regions except HG ($p < 0.001$ in PRD, WCVI, and SOG; $p = 0.037$ in CC), affirming the key role of recruitment in determining subsequent spawning biomass. Both pathways were characterized by narrow posterior distributions, indicating strong support and low uncertainty.

Recruitment was also positively influenced by body condition 2 yr earlier, as indicated by the eLWRF_to_R pathway. This effect was positive across all stocks and statistically significant in WCVI ($p < 0.001$), with marginal significance in CC ($p = 0.096$) and SOG ($p = 0.086$). Feedback effects from spawning biomass to body condition (SBc_to_eLWRF) were positive for all stocks, and significant in CC ($p = 0.03$), implying that higher spawning biomass in earlier years may be associated with improved body condition in subsequent generations.

Predator-related effects on spawning biomass were region dependent. The HSeal_to_SB pathway was negative in HG and WCVI, reaching significance in WCVI ($p = 0.03$), but was significantly positive in PRD ($p = 0.002$), suggesting spatial variability in predator–prey interactions. The GFS_to_SB pathway showed a strong spatial contrast, with positive and significant effects in HG ($p < 0.001$) and CC ($p = 0.026$), but negative and significant in WCVI ($p < 0.001$) and SOG ($p = 0.049$). These contrasting patterns may reflect regional differences in predatory fish composition or feeding behavior. Fishery effects on spawning biomass were also stock specific. Catch had a neg-



(Fig. 5 continued on next page)

Fig. 5. Pathway coefficients (values along arrows) from the best-performing DSEMs for the 5 Pacific herring stocks: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), West Coast of Vancouver Island (WCVI), and Strait of Georgia (SOG). Numbers in parentheses indicate time lags (in years). SBc denotes spawning biomass (SB) when included as a predictor in feedback pathways, and is labeled separately for clarity. Names and descriptions of the drivers and herring response indicators associated with these pathways are provided in Table S1. Arrows indicate the direction and strength of each pathway, with arrow thickness representing the magnitude of effect and colors denoting direction (blue: positive; red: negative). Correlation coefficients are shown for each line, with significance shown in Fig. 6

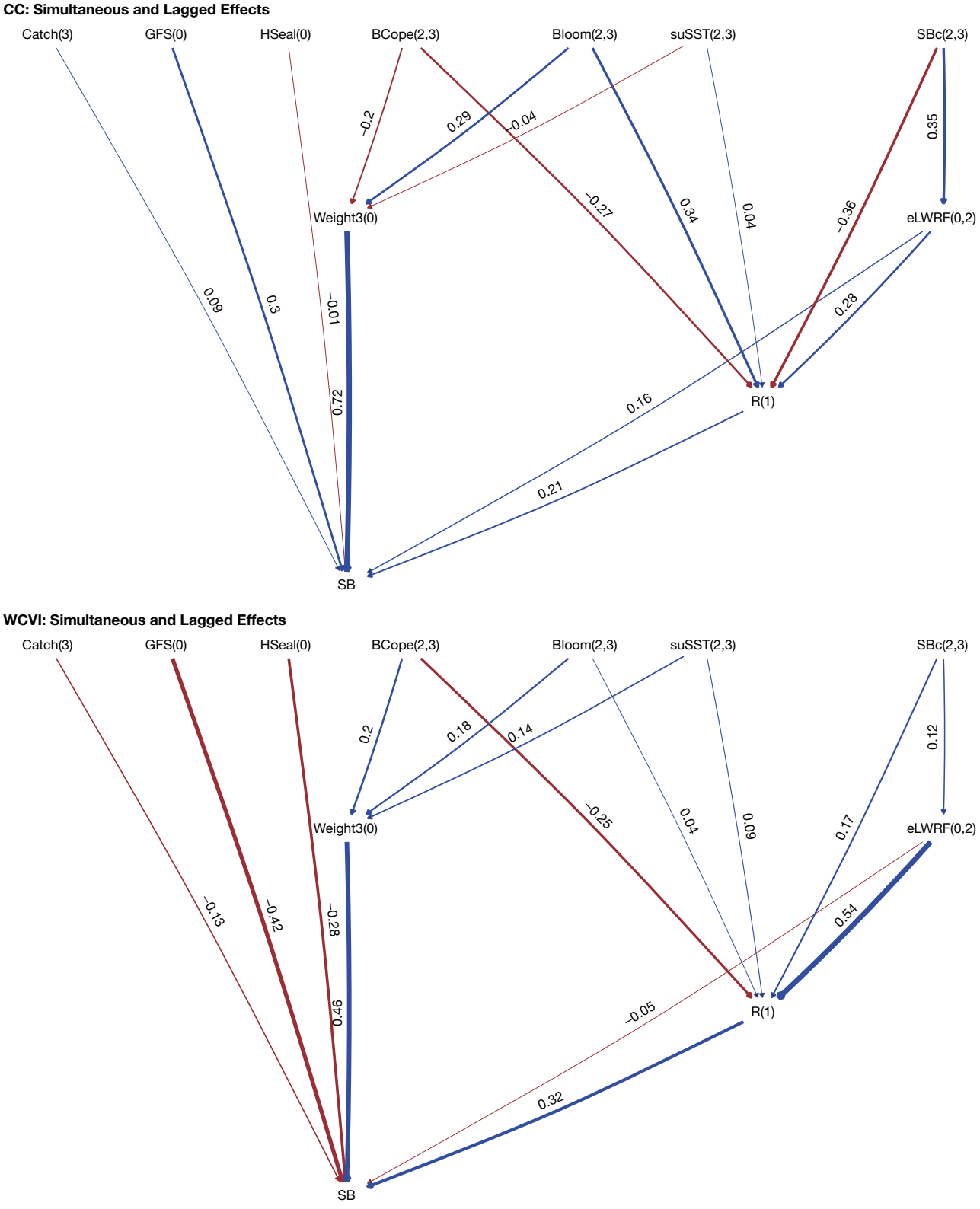


Fig. 5. (continued)

ative effect on spawning biomass in all stocks except CC and was statistically significant in HG ($p = 0.005$), highlighting a stronger fishery impact in this region.

Several prey-related pathways also showed stock-specific significance. BCope positively influenced R in the northern stocks (HG and PRD), but had neg-

(Fig. 5 continued on next page)

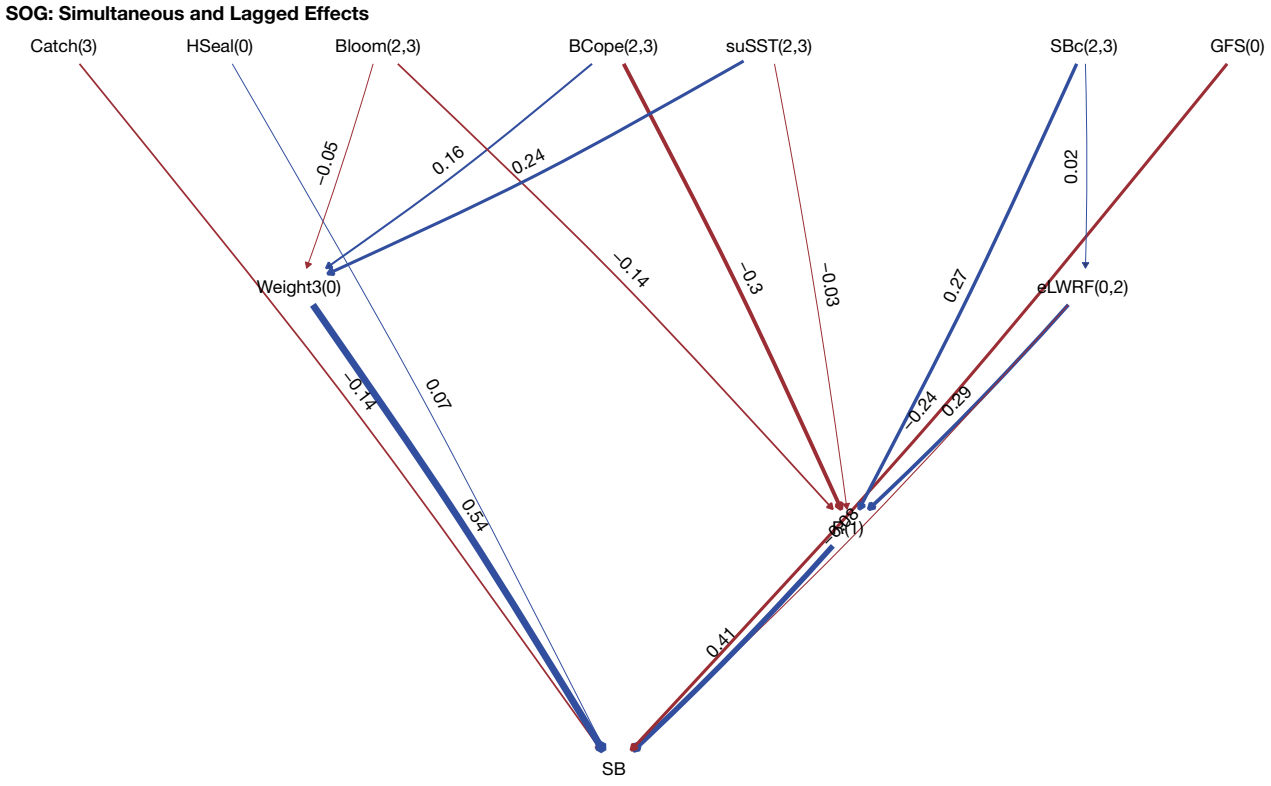


Fig. 5. (continued)

ative effects in WCVI ($p = 0.08$) and SOG ($p = 0.053$). The BCope_to_Weight3 pathway was positive in all stocks except CC, with significance in PRD ($p = 0.048$) and marginal significance in HG ($p = 0.052$), suggesting that boreal copepods support herring growth. In contrast, the effects of suSST on both recruitment (suSST_to_R) and weight at age 3 (suSST_to_Weight3) were weak, inconsistent, and statistically insignificant across all stocks.

Across all stocks, the MCMC chains for the optimal DSEM scenario show good mixing and convergence (results not shown), with effective sample sizes generally exceeding 1000 (Fig. S3) and R-hat values clustering near 1.0 (Fig. S3). These diagnostics indicate that the posterior distributions summarized in Fig. 6 reliably capture parameter uncertainty.

4. DISCUSSION

4.1. Advancing ecological understanding through DSEM and randomized driver exploration

A major innovation of this study was the application of a DSEM framework combined with random sampling of 30 physical, biological, and fishery-related drivers to systematically explore the factors structuring herring dynamics across 5 major BC stocks. Unlike some previous studies on the BC herring stocks that relied on a small, fixed set of drivers, such as that of Xu et al. (2019), which focused exclusively on physical indicators like the PDO, NPGO, upwelling (Up), and downwelling (Down), our broader randomized approach uncovered a far richer and more ecologically diverse set of influential drivers. Notably, NPGO, Up, and Down, the core drivers emphasized by Xu et al. (2019), tended to occur in DSEM scenarios with the highest AIC values in our analysis, suggesting limited explanatory power and, in some cases, a detrimental effect on model performance. In contrast, biological indicators such as Bloom, Chl, BCope, GFS, and HSeal emerged as far more important, repeatedly appearing in the top-performing DSEM scenarios and providing stronger explanatory value for recruitment and spawning biomass dynamics.

This contrast highlights a key strength of our modeling approach: by randomly sampling from a broad suite of candidate drivers, we moved beyond assumptions of large-scale physical dominance and identified more direct biological and ecological mechanisms shaping herring populations. While physical drivers remain ecologically important, affecting, for example, egg development, larval yolk utilization, and spawn timing (Alderdice & Hourston 1985, Rooper et al. 2024), our re-

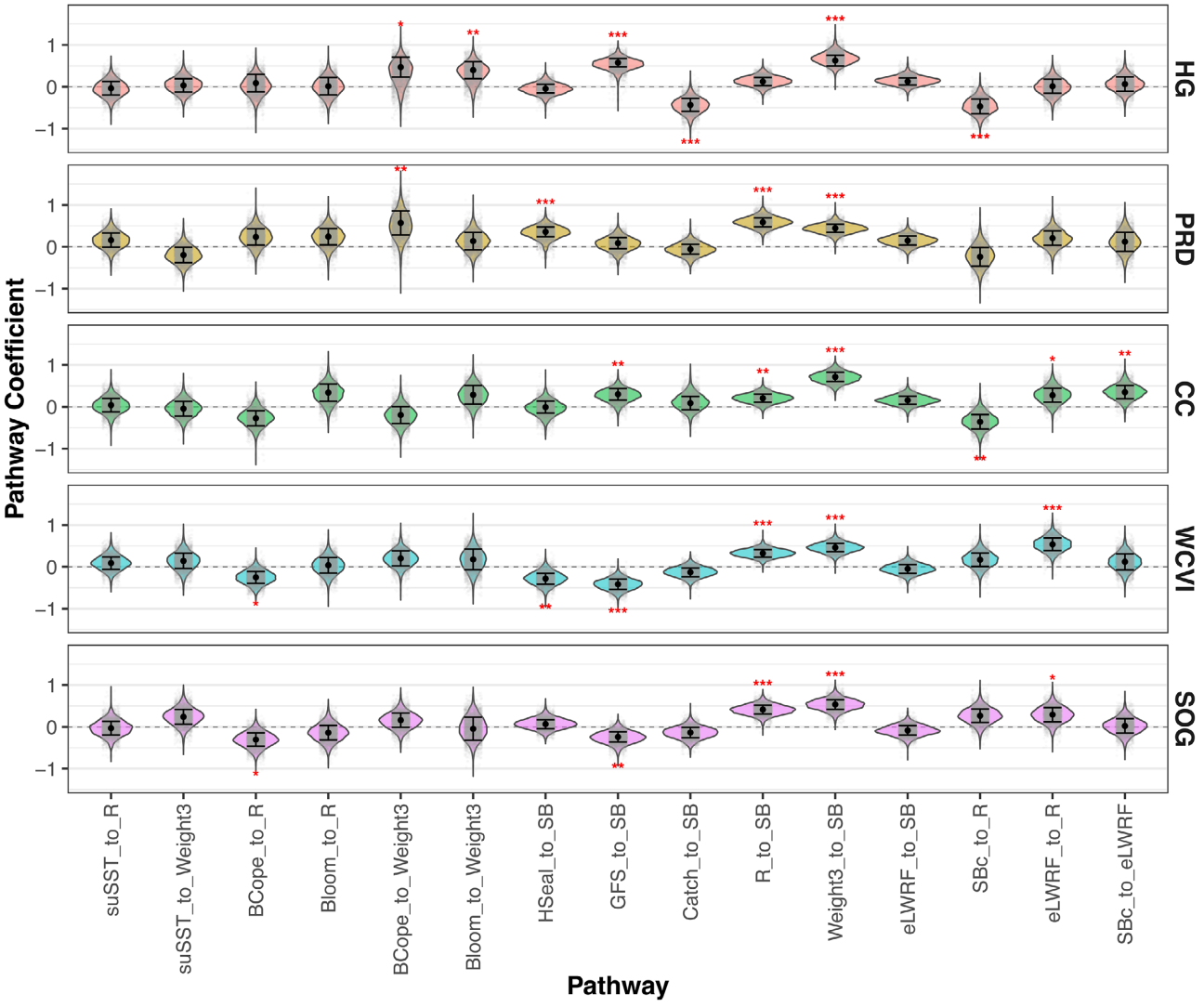


Fig. 6. Pathway coefficients from the best-performing DSEMs for the 5 Pacific herring stocks: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), West Coast of Vancouver Island (WCVI), and Strait of Georgia (SOG). The violins display the full posterior distributions based on the Markov chain Monte Carlo samples, with lightly jittered raw draws shown to illustrate the underlying variation. Posterior means and standard errors are overlaid, and statistical significance is indicated by red asterisks (***$p < 0.01$,**$p < 0.05$, *$p < 0.1$). Names and descriptions of the drivers and herring response indicators corresponding to these pathways are provided in Table S1

sults suggest that, for the period 1990–2024, detailed prey and predator drivers provide stronger explanatory power than coarse climate indices alone. Their stronger performance in the DSEM likely reflects their closer mechanistic linkage to recruitment, growth, and survival, even though they remain embedded within the larger physical oceanographic context.

These findings are broadly consistent with other previous research on Pacific herring in BC. For example, the timing of prey availability can affect the abundance of herring (Schweigert et al. 2013, Boldt et al. 2018), and herring productivity has been correlated with primary production (Ware & Thomson 2005, Perry & Schweigert 2008). Predation has also been identified as a driver of herring dynamics, although the importance of different predators (groundfish or marine mammals) varies among studies (e.g. Ware & McFarlane 1986, Schweigert et al. 2010, Godefroid et al. 2019, Doherty et al. 2025).

Similarly, Boldt et al. (2025) expanded the set of considered 'pressures' on Haida Gwaii herring to include both physical and biological indicators, using well-established modeling techniques such as gradient forests and dynamic factor analysis. Their work provided valuable insights into the roles of individual drivers like predator abundance and upwelling trends, particu-

larly when applied to targeted subsets of variables. However, their analysis focused on fixed sets of drivers and primarily evaluated simultaneous relationships. In contrast, our DSEM framework integrated ecologically informed time lags, guided by species life history, allowing us to represent biologically plausible delayed effects. Such incorporation of lagged pathways improves ecological realism compared to simultaneous-only models and underscores the importance of temporal structure in fish population modeling.

It is worth noting that the recruitment data correspond to cohorts that have already survived their early life stages; therefore, physical and prey drivers were assigned a 2 yr lag to approximate the time between environmental conditions experienced during the egg–larval–juvenile period and the year in which that cohort is enumerated as recruitment in the assessment. We acknowledge, however, that environmental influences often operate over much shorter biological windows (e.g. weeks to months). Thus, we do not exclude the possibility that alternative or distributed lag structures could more accurately represent continuous environmental forcing, such as marine heatwaves or prey mismatches occurring only months before recruitment (Cushing 1969, Schweigert et al. 2013). For this reason, the fixed lags used here should be viewed as simplifications aligned with the stock assessment data structure rather than tests of alternative temporal configurations.

We further note that predation on recruits was not explicitly modeled as a separate lagged pathway, because natural mortality prior to recruitment age is already embedded within the stock assessment recruitment estimates. This simplification is consistent with other ecosystem-based analyses of herring but could be improved in future work through incorporation of juvenile survey data or seasonal predator–prey time series. Future extensions of the DSEM framework may therefore include explicit estimation of lag structures, the use of distributed lag formulations, and the evaluation of seasonal windows of sensitivity. Together, these approaches would enhance the temporal realism of driver–response relationships. Such developments would allow a more nuanced assessment of how short-term environmental perturbations propagate through the herring life cycle, an increasingly important consideration under contemporary climate variability.

4.2. Regional comparisons and contrasts in key drivers across stocks

Our randomized DSEM demonstrated some regional contrasts in the driver portfolios associated with herring response indicators. The eLWRF_to_R and SBc_to_eLWRF pathways were consistently positive across stocks, highlighting potential feedbacks linking female body condition, reproductive output, and future cohort strength. Positive maternal-condition effects have been documented in other fish species, where higher female energy reserves improve egg quality, larval survival, and early juvenile performance (Berkeley et al. 2004). Positive relationships between herring length and both ovary weight and gonadosomatic index indicates that larger herring have increased reproductive output (Hay et al. 2019). However, the magnitude of these pathways varied among stocks, with weaker effects in some regions, likely reflecting differences in prey availability, body-condition dynamics, and uncertainty in stock-specific estimates.

The SBc_to_eLWRF pathway may reflect circumstances in which years of higher spawning biomass coincide with improved ecosystem conditions, such as enhanced prey field, or a greater proportion of older, higher-condition individuals in the spawning population, rather than direct density-dependent regulation. Comparable positive associations have been reported for Atlantic herring and other clupeids when environmental conditions simultaneously benefit both adults and their prey fields (Engelhard & Heino 2004, Husebø et al. 2007). The observed variability among stocks is consistent with evidence that herring condition responds to regional prey communities, temperature regimes, and energetic constraints (Mackas et al. 2007).

Among the predation-related drivers, GFS was consistently negative for SB in the southern stocks (WCVI, SOG) but neutral or positive in HG, PRD, and CC, pointing to regional differences in predator composition and spatiotemporal overlap with herring. Such spatially differentiated predator effects are consistent with previous work showing that predator–prey overlap and local predator assemblages can strongly mediate herring mortality (Schweigert et al. 2010), particularly on the WCVI shelf where hake–herring interactions are well documented (Ware & McFarlane 1986, Godefroid et al. 2019). The driver Hake exhibited a strong southward signal: it ranked among the second or third lowest AIC drivers in CC, WCVI, and SOG, yet was associated with substantially higher AIC in HG and PRD, consistent with a south-skewed distribution and seasonal residency. This pattern aligns with known biogeographic structure in hake abundance, suggesting that predator pressure is more influential for southern herring stocks, a result broadly consistent with studies of small pela-

gic fishes where predation risk often varies with latitude (Alheit & Peck 2019).

For HSeal, its effects on spawning biomass were negative in HG and WCVI but positive in PRD, suggesting that local prey fields, predator behavior, and habitat structure mediate predation outcomes. One plausible hypothesis is that in regions such as PRD, higher seal abundance co-occurs with periods of abundant alternative prey (e.g. salmon smolts, hake, groundfish), reducing predation pressure on herring and resulting in a positive association between HSeal and SB (Olesiuk 1993, Thomas et al. 2017). Another possibility is that seals aggregate in areas of higher overall prey productivity, creating a positive correlation with herring biomass through numerical responses rather than increased predation (Smout & Lindstrøm 2007). These mechanisms are consistent with the idea that predator effects on forage fish are context-dependent and mediated by regional differences in prey communities and predator foraging strategies. Among fishery-related drivers, Catch generally exhibited negative effects on spawning biomass, whereas Bycatch showed contrasting associations across regions. In WCVI and SOG, the positive Bycatch_to_SB pathways warrant cautious interpretation and are unlikely to represent a direct positive effect of bycatch on spawning biomass. Instead, these associations may reflect covariance with periods of higher stock abundance, spatial and temporal overlap between fishing activity and herring distribution, or changes in fishing practices and targeting behavior, as commonly discussed in fisheries assessment literature (Maunder & Punt 2004).

Bottom-up pathways also differed by region. For example, copepod biomass (e.g. BCope) tended to support R in HG and PRD but was negative in CC, WCVI, and SOG, implying regional variation in trophic linkage strength, prey quality, or phenology. This north–south divide mirrors well-established gradients in zooplankton community composition, where northern regions are dominated by larger, lipid-rich boreal copepods known to improve forage quality for juvenile herring (Mackas et al. 2007). In addition, such negative associations can occur even when boreal copepods are high-quality prey, because regional dynamics may produce prey–larval timing mismatches (Mackas et al. 2007, Schweigert et al. 2013), shifts in zooplankton community composition that reduce availability of smaller prey for first-feeding larvae (Keister et al. 2011), or oceanographic conditions that favor boreal copepods while simultaneously reducing larval retention or early survival (Ware & Thomson 2005).

Seasonal SST effects on Weight3 skewed positive in WCVI and SOG and negative in PRD, with HG and CC mixed, indicating region-specific or latitudinal thermal windows; warming may enhance early survival where prey timing aligns, but may increase mismatch or metabolic costs elsewhere. Such stock-specific sensitivity to thermal conditions has been observed in other herring systems, including Norwegian spring-spawning herring where temperature-dependent processes during early life stages influence recruitment variability (Vikebø et al. 2010), and in small pelagic fish globally, where regional thermal regimes often produce contrasting recruitment responses (Peck et al. 2021).

Taken together, these contrasts indicated that herring stocks were regulated to some extent by region-specific combinations of top-down predation, bottom-up prey dynamics, and physical forcing, motivating targeted research into stock-specific mechanisms. Accordingly, management strategies and rebuilding expectations should reflect regional driver portfolios — incorporating spatial predator–prey overlap metrics, testing for depensatory predation where predator aggregation is likely, and emphasizing prey phenology and thermal windows where bottom-up control dominates.

These regional contrasts are also broadly consistent with ecological patterns documented for herring and other small pelagic fish species, which commonly exhibit spatially structured responses to predators, prey regimes, and climate variability (Alheit & Peck 2019, Peck et al. 2021). The parallels between our findings and prior studies underscore that the spatial heterogeneity revealed by the DSEM approach reflects real ecological mechanisms rather than statistical artefacts, and highlight the value of explicitly accounting for regional ecological processes when assessing stock dynamics.

4.3. Group-based random sampling to ensure balanced and interpretable DSEM configurations

A central strength of our modeling framework is the structured, group-based random sampling approach, which directly shaped the ecological insights emerging from the DSEM rather than serving solely as a model-building procedure. By allowing thousands of biologically plausible driver combinations to compete, the framework follows the principles of information-theoretic multi-model inference (Burnham & Anderson 2002), ensuring that influential drivers reflect consistent signals across many alternative

hypotheses rather than dependence on any single researcher-defined model. This approach preserves the ecological identity of each driver, avoiding the interpretability limitations associated with dimension-reduction techniques such as dynamic factor analysis or principal component analysis (Jolliffe & Cadima 2016). The DSEM further retains explicit causal pathways and temporal structure (Thorson et al. 2024), enabling clearer interpretation of how individual drivers influence recruitment, body condition, and spawning biomass.

Grouping correlated drivers and selecting one representative from each group minimized multicollinearity while still capturing the breadth of potential ecological mechanisms. This design parallels ensemble-based ecological modeling strategies used in fisheries and ecosystem science (e.g. Fulton et al. 2011, Planque 2016), where broad exploration of the driver space helps distinguish robust mechanisms from model-specific artifacts. In our context, it enabled systematic evaluation of alternative representations of physical, biological, and fishery-related drivers, highlighting which drivers consistently improved model performance. The approach is flexible and scalable: the DSEM framework does not require a 6-driver configuration, and increasing the number of drivers per scenario is feasible, though at the cost of greater model complexity and potential overfitting.

Overall, the group-based sampling framework strengthens inference by ensuring that key mechanisms are supported across many competing models, not just a narrow set of hypotheses. Although the driver set used in this study is broad and ecologically informed, we acknowledge that it is not exhaustive; additional pressures such as coastal modification or vessel disturbance may also influence herring dynamics and could vary in importance over time (Penttila 2007, Williams et al. 2015). The framework expands ecosystem-based inference beyond previous herring studies that relied on fixed covariate sets or dimension-reduction techniques (e.g. Xu et al. 2019, Boldt et al. 2025), by evaluating a larger suite of ecological mechanisms within a unified structural equation framework.

4.4. Addressing multicollinearity without dimension reduction: strength of the sampling design

A particularly valuable feature of this framework is its ability to address multicollinearity while retaining the ecological meaning of individual drivers. By grouping highly correlated drivers and sampling only one from each group per model, the framework prevents the co-occurrence of collinear predictors without needing to reduce dimensionality or drop biologically meaningful variables. In one special case, GFS and SSLion, correlated only in WCVI and SOG, were assigned to separate groups, allowing co-occurrence in 233 of 6098 DSEM scenarios (3.8%). Given this limited incidence, any bias from their occasional joint inclusion was expected to be negligible. Importantly, retaining all original drivers while preventing multicollinearity ensures that each variable had a fair chance of inclusion across scenarios, supports statistically stable coefficient estimation, and avoids arbitrary pre-selection of drivers.

Moreover, this approach allowed us to evaluate not just individual driver effects within and across stocks, but to also examine the comparative importance of alternative drivers within the same group. For example, substitution experiments showed that replacing top-performing drivers with weaker alternatives from the same group consistently reduced model performance, reinforcing both the effectiveness of the sampling design and the ecological relevance of the retained drivers.

Overall, this design supports robust inference in high-dimensional ecological settings by managing collinearity through sampling design rather than data simplification, making it especially suited for species like herring where many drivers operate simultaneously across physical, biological, and anthropogenic domains. This is particularly advantageous for fisheries ecology, where mechanisms such as match–mismatch dynamics (Cushing 1969), prey quality shifts (Mackas et al. 1998), and predator–prey overlap (Schweigert et al. 2010) are best represented by explicit variables rather than latent components. The approach is also aligned with statistical guidance emphasizing careful treatment of collinearity while retaining mechanistically important predictors (Graham 2003). By integrating these principles within a structural equation modeling framework, our implementation contributes to ecosystem-based approaches to fisheries management and emerging causal modeling frameworks (Thorson et al. 2024).

4.5. Future directions for long-term analysis

While recruitment and spawning biomass data for herring are available from as early as 1951, this study focused on the period 1990–2024 to ensure the inclusion of ecologically important biological drivers. Although longer time series can improve statistical

power, relying solely on physical and fishery drivers available in earlier decades risks producing biased models that overemphasize environmental forcing while omitting critical bottom-up mechanisms. Our decision to prioritize ecological completeness over historical length was intended to preserve the interpretability and biological realism of the driver–response relationships. This trade-off between temporal coverage and ecological completeness is a common challenge in small pelagic and forage-fish research (Lluch-Belda et al. 1992, Alheit & Peck 2019, Boldt et al. 2022), and our DSEM implementation offers a structured way to make this choice explicit.

Nonetheless, extending the modeling framework to the full 1951–2024 window remains an important avenue for future work. The ability of the DSEM framework to accommodate missing data enables the inclusion of biological variables with incomplete records, allowing earlier periods to be modeled primarily with physical and fishery-related indicators while maintaining the potential to incorporate biological effects when data permit. This approach aligns with best practices in ecosystem modeling, which emphasize flexibility and transparency when data completeness varies across variables and timeframes.

Moreover, broadening the temporal scope offers a valuable opportunity to examine how herring dynamics have responded to major North Pacific climate regime shifts, such as those in 1976, 1989, and 1998, which are well-documented to have restructured marine ecosystems (e.g. Hare & Mantua 2000). These shifts may have altered the strength or even the direction of key ecological linkages. As demonstrated in salmon systems (e.g. Litzow et al. 2018), traditional regression models assuming stationary relationships can obscure important temporal variability in climate–biological interactions. Extending DSEMs to incorporate time-varying coefficients or regime-by-driver interaction terms would allow explicit testing of whether driver effects on recruitment, growth, and spawning biomass shifted across these climatic transitions, similar to patterns described for other small pelagic species undergoing climate-driven reorganization (Alheit & Peck 2019).

By incorporating time-varying coefficients or regime-based interaction terms within the DSEM framework, future analysis can explicitly evaluate how the importance of different drivers changes across climate regimes. Similar non-stationary responses have been documented for small pelagic fishes in other regions (Lluch-Belda et al. 1992, Alheit & Peck 2019), suggesting that a regime-aware DSEM implementation for herring would also be broadly relevant to understanding climate-contingent driver portfolios in other forage-fish systems.

Incorporating these enhancements would support a more nuanced understanding of long-term variability in herring population dynamics, strengthen the detection of climate-contingent relationships, and improve the predictive capacity of ecosystem-based models under ongoing and future environmental change. Because the DSEM framework is generic, these long-term, regime-focused extensions could be directly transferred to other small pelagic and demersal species where multi-decadal data are available, further increasing the applicability of the approach for ecosystem-based fisheries management.

5. CONCLUSIONS

This study introduced a novel application of a randomized DSEM framework to disentangle the complex drivers of Pacific herring dynamics across 5 major stocks in BC. By systematically sampling from a broad suite of 30 physical, biological, and fishery-related drivers, and evaluating over 6000 model scenarios, we were able to identify consistent and biologically interpretable pathways that structure recruitment, growth, and spawning biomass.

Our results highlight the dominant role of intrinsic stock dynamics, particularly the positive effects of recruitment and age-3 weight on spawning biomass, and a strong feedback loop between spawning biomass and female body condition. Among external drivers, biological drivers such as spring bloom timing, copepod biomass, and predator biomass were far more influential than traditionally emphasized physical variables like downwelling or upwelling indices. This finding underscores the importance of bottom-up trophic controls and localized ecological mechanisms in shaping herring population trajectories.

Importantly, our modeling framework addressed multicollinearity and high dimensionality without discarding informative variables or relying on dimension-reduction techniques. The group-based random sampling design preserved ecological interpretability while maintaining statistical rigor and fair model comparisons.

Together, these findings provide new insights into stock-specific and coastwide regulation of Pacific herring and reinforce the value of randomized, multi-driver modeling approaches for ecosystem-based approaches to fisheries management. This work lays the foundation for future extensions that incorporate longer time series, climate regime shifts, and non-stationary processes, and for applying this modeling

framework to other small pelagic species and fish stocks more broadly, thereby improving our ability to anticipate population responses to ongoing environmental change.

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